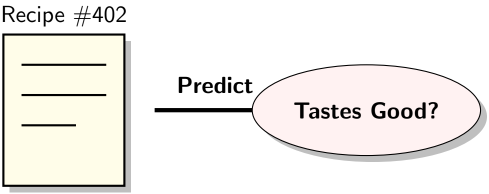
The Limits of Prediction

From Baking Cakes to Rice's Theorem

Theory of Computation

A Judge With No Time

Imagine a world-famous baking competition with thousands of contestants. The Head Judge is completely exhausted and makes a seemingly brilliant request to her assistants.



She says: *"I don't want to bake all these cakes. I want you to read the written recipes and tell me, with 100% certainty, if the final cake will taste sweet."*

The Trap of Complex Instructions

Why Reading Isn't Enough

It is easy to scan a recipe and say, "Yes, this contains the word 'sugar'." That is a superficial property of the paper. But predicting the *experience* of the cake is different.

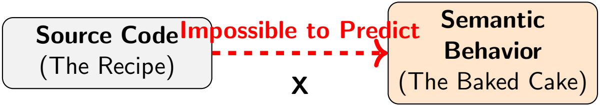
The Paradox of Action

What if a recipe says: *"Taste the batter. If it's too thick, add lemon. If it foams, add salt. Keep doing this until it feels right."* You cannot predict the outcome just by reading, because the instructions change based on the physical act of baking!

The Reveal: What is Rice's Theorem?

Connecting the Story to Computer Science

The exhausted judge trying to avoid baking is exactly what computer scientists try to do when writing "Static Analysis" tools. We want a master program that reads other programs' code and predicts their behavior without running them.



Undecidability

Henry Gordon Rice proved mathematically that **all non-trivial semantic properties of programs are undecidable.**

You cannot write a perfect tool that reads code and predicts its ultimate behavior. To know what the code truly does, you cannot just read the paper—you have to "bake the cake" and execute the program.